
Stroke Risk Prediction Project

Assignment 3
Final Report Document

Team members (Group 6)

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PROJECT MANAGEMENT

1. Contributions of each member of the group

Pamela: I contributed to the project by proposing the team name and actively participating in the development of the team rules, project charter, and final report. My primary focus was on the dashboard design, where I worked on the visual layout and color scheme to enhance user experience. Additionally, I was responsible for planning and implementing the usability testing, including selecting the System Usability Scale (SUS) and conducting think-aloud interviews. I also collected and analyzed the results from both methods to inform design improvements.

Htet: I was actively involved in all phases of the project. Regarding preparation documents, I contributed to several key sections, including the Team Rules document, the Project Charter, and the final reports. This work included write-ups and development on low-fidelity prototypes, team structure and role descriptions, stakeholder analysis, high-level alternatives, and essential annexes such as the Work Breakdown Structure (WBS) and the risk management tools. In the design phase, I took responsibility for developing the Risk Prediction and Preventive pages. Relating to these pages, I participated in model testing, focusing on parameter tuning and applying relevant methods.

Pardon: I participated in the development of the team rules, project charter and the final report. I was also giving inspiration to the dashboard design, contributing to some coding and assisting where help was needed. In particular, I coded and developed certain features and graphs on the About and Descriptive pages.

Chantale: I focused mainly on the writing, specifically the project charter and the final project management report. I also contributed to the dashboard by providing initial graphs and analysis on the Descriptive page.

Johannes: My contribution to the project was included in several areas. I worked both on the document and on coding. I was responsible for performing data preprocessing to enhance data quality and consistency. I also attempted to train the model and took part in writing in our document throughout the development process, as well as actively participating during our discussions throughout the development time.

2. Team Development Experiences During the Project based on Tuckman's model

The **Tuckman's model (1)** best describes the stages of group development experienced during our project. Below is the group development of Neuroinsight group

Forming:

In the first week, we defined the project scope, assigned tasks, and created a GitHub repository. Everyone was polite and cooperative but somewhat unsure of their specific roles. We also agreed to hold weekly update meetings via Zoom to stay aligned. A google document was created for documentation so group members could work parallel.

Storming:

During the coding phase, the group experienced some challenges since the database was imbalanced and it was difficult to validate the model created. Most issues were resolved through brainstorming sessions, leading to joint decisions on coding approaches and data handling.

Norming:

By the time we reached the dashboard development stage, our roles and communication patterns became much clearer. Weekly progress updates helped maintain accountability, and collaboration felt more structured and efficient.

Performing:

Near the end of the project, teamwork was highly effective. Members worked semi-independently on their assigned tasks but stayed synchronized through GitHub pull requests and regular check-ins. Those with stronger coding skills focused on programming, while others contributed to writing and documentation, leveraging each person's strengths.

Adjourning:

The final report writing stage brought the team together once again to reflect on our process, finalize the deliverables, and formally close the project.

3. Team Member Experiences: Comfort, Challenge, and Personalities

Comfort

The team's roles were not strictly defined at the beginning, which allowed members to naturally gravitate toward the tasks they felt most comfortable with. For example, some focused on coding while others contributed more to documentation or visualization. Despite this flexibility, collaboration remained strong; members regularly consulted and supported each other when challenges arose.

Challenge

Several members felt challenged when taking on unfamiliar tasks, such as learning Streamlit, debugging code, or writing technical documentation. These challenges, although initially stressful, created valuable learning opportunities and encouraged knowledge sharing within the group. The willingness to step outside of one's comfort zone contributed to both personal and team growth.

Relation to Personalities

The comfort and challenge levels appeared closely linked to each member's personality and background. Those with strong analytical and logical thinking skills were naturally drawn to data preprocessing and modeling, while members with creative tendencies preferred dashboard design and presentation. The project environment allowed individuals to leverage their strengths while also exploring new areas, promoting both confidence and skill development.

4. Leadership styles and reflections of team members

Pamela: I would describe my leadership style as a blend of Steadiness and Compliance, which reflects my patient, thoughtful, and cooperative nature. I strive to create a safe and inclusive environment where every team member feels that their opinions and contributions are valued and respected. I particularly enjoyed contributing to both the writing and technical aspects of the project. I also took responsibility for ensuring coherence and consistency in the written materials we produced. I regularly reviewed my teammates' work, offering suggestions for improvement when appropriate, while still respecting their autonomy. I trusted my team to meet our agreed deadlines, making collaboration smooth and productive.

Htet: My leadership style reflects a Compliance–Steadiness (C/S) profile. I am a reliable and supportive leader who places high value on accuracy, quality, well-structured work, high standards, and collective success. I always try to create a stable, productive environment where all team members feel valued, supported, and productive. I ensure that our team's work products are delivered with the highest quality possible within the given timeframe.

Johannes: My leadership style would be described as inclusive, trying to give ideas and hearing other people's views and thoughts. The key factor is to make sure everyone is comfortable speaking and sharing their ideas. During this project I was jumping between coding and writing in the document. Working wherever I was needed at that time.

Pardon: I am a leader who wants to give other members their creative freedom. I do participate by doing my part and participating in collaborative assignments. I support where I can and ask for help when I need it. I am not a multi-tasker and I prefer tasks to be handled sequentially.

Chantale: I would describe my leadership style as supportive and participative. I enjoy encouraging and lifting up others, ensuring that everyone feels valued and motivated within the team. I am good at planning my time and staying organized, which helped me manage my contributions effectively throughout the project. Although I am stronger in writing and documentation than in coding, I took the initiative to create my own small repository to practice coding and improve my technical skills. This reflects my willingness to learn and lead by example through continuous self-development. I believe good leadership involves creating a positive environment where everyone can contribute according to their strengths while still being challenged to grow.

Joint Reflection: Leadership was shared depending on the task. No single member dominated leadership.

Regarding the leadership assignment and group development in general, what worked well? What did not? And what could have been done differently and how?

What worked well:

There was good chemistry within our group, and communication flowed easily. We reached agreements without conflict, and everyone contributed according to their strengths. Those who felt comfortable with certain parts of the assignment took the lead in those areas, while others offered support when needed. Collaboration and mutual respect were strong throughout the project, which helped maintain motivation and trust. We set deadlines after every meeting and members followed them. Clear GitHub workflow worked well, regular updates, task sharing, and mutual support when someone faced technical difficulties and overall good team chemistry.

What did not work well

Although overall teamwork was smooth and collaborative, the roles were not clearly defined at the outset, which led to some initial ambiguity in task allocation. The team is all from medical background who do not have high technical and coding skills. We experienced a lot of challenges in developing the dashboard, for some debugging issues like bleeds in the dashboard, we could not fully address it although we tried our best with recommended methods from literature and AI tools. A team with mixed medical and technical backgrounds would tackle those problems better,

What could have been done differently

At the start of the project, we could have clarified individual roles and set more specific goals for each member. Having a short-written plan outlining responsibilities and deadlines might have improved coordination and ensured a more even contribution. Regularly rotating tasks could also have helped everyone gain experience in both technical and non-technical aspects of the project.

5. Development Approach and Its Impact on the Project

The NeuroInsight team followed an Agile-approach characterized by iterative development, small task divisions, and frequent check-ins. We organized our work in short cycles, where each member completed a specific part of the project and then shared progress during weekly meetings or through GitHub updates. We also tested and refined the code continuously to ensure functionality and to detect errors early.

The team adopted an Agile approach, emphasizing iterative progress, collaboration, and adaptability. The work was divided into smaller, manageable tasks that were continuously reviewed and refined. Regular check-ins through Zoom meetings and GitHub updates facilitated

transparency and early identification of issues. Continuous integration and testing ensured that each component of the project was functional before proceeding to the next stage. Although a fully structured Agile framework (such as Scrum) was not implemented, the team's working style reflected key Agile principles, including flexibility, collective ownership, and incremental improvement.

The application of this approach had predominantly positive effects on both the project process and outcome. On the process level, it enhanced communication, accountability, and adaptability, allowing team members to learn from one another and adjust quickly to challenges. The iterative structure supported continuous feedback and problem-solving, contributing to efficient task management and a collaborative team environment. On the product level, the approach improved the quality and functionality of the dashboard by enabling early testing and gradual refinement. However, the absence of clearly defined roles occasionally led to overlapping efforts and uneven task distribution. Overall, the Agile-inspired method proved effective and well-suited to the project's scale and learning objectives.

End Product

The final outcome of the project is a functional and user-friendly dashboard designed to visualize and interpret data from the Stroke Prediction dataset. The dashboard effectively integrated key analytical features, allowing users to explore relationships between variables and gain insights into stroke risk factors. It demonstrates solid technical functionality and meets the team's initial project objectives.

On Process:

On the **process side**, the project fostered significant learning, collaboration, and mutual support within the team. Members developed both technical and communication skills through continuous engagement and feedback. However, the process could have benefited from a more structured sprint planning approach to enhance coordination and time management. This was particularly challenging as one team member had limited time, which occasionally limited full participation. Despite these minor constraints, the overall outcome was successful, reflecting strong teamwork and commitment.

Think about a hypothetical scenario where you could restart the project: would you have chosen the same project development methodology? Explain your answer.

If we were to restart the project, we would likely still choose an Agile-inspired methodology, but we would apply it in a more structured way. Agile suited our small group size and allowed us to adapt quickly to challenges, such as switching models when logistic regression underperformed. However, we did not always follow clear sprint planning or task prioritization, which sometimes

led to duplicated work and inefficiencies. A more formalized Agile approach (e.g., defining weekly sprints with deliverables) would likely have improved coordination.

6. Lessons learned on the Agile approach

What worked well: Because Agile emphasizes adaptability and collaboration, the team was able to reassign tasks and responsibilities efficiently when necessary. For example, members who initially focused on coding could switch to documentation or testing as project needs evolved. This adaptability fostered learning, maintained engagement, and ensured that no part of the project was neglected. Iterative testing and regular feedback loops allowed us to identify problems early and improve both the model and dashboard. This flexibility kept the project moving despite technical challenges.

What could be done differently: In future projects, it would be beneficial to establish clearer **coding standards** and adopt task-tracking tools (such as Trello or GitHub Projects) early in the process. These tools would help coordinate activities, reduce duplicated efforts, and provide better visibility of each member's progress. Additionally, maintaining more consistent documentation of decisions and workflow updates would improve continuity and make it easier for team members to adapt when roles or responsibilities change.

7. General feelings among the group members Tomas-Kilman conflict model

The Thomas–Kilmann Conflict Mode Instrument (TKI) (2) is a framework developed by Kenneth W. Thomas and Ralph H. Kilmann to assess how individuals respond to conflict situations. The model identifies five conflict-handling styles based on two dimensions: **assertiveness** (concern for self) and **cooperativeness** (concern for others). The five styles are: competing, collaborating, compromising, avoiding, and accommodating.

Our team experienced very few, if any, conflicts throughout the project. Communication was open and respectful, and members were supportive, and everyone's voice was heard. According to the TKI, this can be interpreted as a tendency toward collaborating and accommodating conflict-handling styles, where individuals prioritize shared goals and harmony over competition. This cooperative dynamic contributed to smooth teamwork, efficient task completion, and a positive group atmosphere.

During our project, no conflict or disagreement occurred that significantly impacted our development. When disagreement arose or different ideas were presented, we had a group discussion to make sure everyone's voices were heard and discussed the pros and cons of that idea. Even when the leadership role was present, we respected our differences and always tried to come up with the best solution that all would be behind.

8. Adherence to the Work Planning and Scheduling Tools

Overall, the team was able to follow the Work Breakdown Structure (WBS) and initial schedule reasonably well. The early phases (dataset exploration, preprocessing, and initial modeling) were completed on time. However, there were delays during dashboard integration and the busy schedule of group members. The flexible, Agile-inspired approach allowed us to adapt to the schedule when needed without significantly affecting the overall progress. Overall, the WBS and Gantt plan provided valuable guidance, even though the team applied them more as a framework than as a strict timeline.

Significant Deviation:

The Gantt chart initially allocated one week for dashboard development, but this stage took a longer time due to our busy schedule. As a result, the reporting phase was compressed, and team members had to work in parallel on documentation and dashboard polishing. Despite this deviation, the team delivered the final product on schedule by reallocating tasks and increasing collaboration near the deadline.

Evidence of the tools used for project management.

The project management and development tools selected proved to be highly effective in supporting collaboration and productivity throughout the project. **GitHub** was essential for version control, collaborative coding, and preventing file overwriting. The use of pull requests enabled smooth coordination of individual contributions. **Streamlit** facilitated rapid development of an interactive dashboard, allowing the team to visualize model outputs efficiently. **Google Docs and Google Drive** supported real-time collaborative writing, document sharing, and resource organization. **Visual Studio Code** served as the primary development environment for coding and debugging. For communication, **Zoom** was used for weekly meetings and progress discussions, while **WhatsApp** provided a quick and informal channel for daily coordination. Together, these tools enhanced transparency, teamwork, and overall project efficiency. Evidence of our WhatsApp group and GitHub group project are attached in the appendix section.

DATA SCIENCE

1. Dataset, problem domain, and intended target users for Neuroinsight dashboard.

The selected dataset was the Stroke Prediction Dataset from Kaggle (3). The dataset contains 5,110 observations with 12 attributes (patient id, gender, age, hypertension, heart_disease, ever_married, work_type, residence_type, avg-glucose-level, bmi, smoking_status and stroke (target: 0 = no stroke, 1 = had stroke). It includes both stroke and non-stroke cases, making it suitable for supervised machine learning classification.

The problem domain is healthcare risk prediction, specifically identifying individuals at higher risk of stroke. Early detection is critical since stroke is a leading cause of death and disability worldwide, and timely intervention can significantly improve outcomes.

The intended target users of the web dashboard are clinicians, public health professionals, and health researchers who need an intuitive tool to visualize stroke risk factors, explore the dataset interactively, and test prediction outcomes without requiring advanced coding knowledge.

2. Processing pipeline

When we received our dataset that would be used for our project, we had to do preprocessing before we could use it to create our model. Preprocessing is important to avoid miscalculations and false predictions and to achieve clarity.

- Importing the CSV file and inspecting it first to find any missing or inconsistent values
- All numerical values were changed to 0 from NaN to include them in the calculations
- One column, “ID”, was dropped due to not being necessary for calculations, as it is only for identification.
- Datatypes of most columns were changed from objective to numerical, categorical or boolean for training
- We also opted to change the names of all the columns for clarity

All these steps were taken to ensure data consistency, avoid noise and prepare the data for analysis.

3. Descriptive Analytics tab

1. What is the gender distribution in the Stroke prediction dataset?
2. What is the percentage of patients with heart disease disaggregated by types of work and urban/rural residency?
3. What is the median age of patients with stroke and without stroke?
4. What is the stroke rate by smoking status? Which has the higher rate?
5. What is the average glucose level of stroke and no-stroke patients?
6. How do the average BMI values vary between patients with hypertension and patients without hypertension?

We conducted a gender analysis using a simple pie chart that illustrated the distribution of participants: males comprised 58.6%, females accounted for 41.4%, and others represented 0.01%. To address the second analytical question, we employed a bar graph that highlighted key insights, revealing the highest proportion (10%) among self-employed individuals living in urban areas, while the lowest proportion (5.6%) was among those with private jobs in urban areas. For the analysis of median age differences between stroke and non-stroke patients, we utilized a box plot. This chart indicated that the median age for stroke patients is lower, at 40, with most individuals

falling within the 20 to 60 age range. In contrast, the median age of non-stroke patients is 65, with a narrower age distribution.

Regarding the stroke rate by smoking status, we selected a simple column chart, showing the rates are higher among former and current smokers, with figures of 7.9% and 5.3% respectively, compared to never smokers, who have a rate of under 4.7%. We conducted an analysis to compare the average glucose levels between stroke and non-stroke patients using a simple bar graph. The results indicated that stroke patients had a significantly higher average glucose level of 132 mg/dL compared to 104 mg/dL in non-stroke patients. To examine the variance in average BMI between the two groups, we utilized a box plot. The box plot revealed that the mean BMI values were relatively close, with stroke patients averaging 29.9 and non-stroke patients averaging 27.5. However, it also highlighted that the range of BMI values was considerably wider among stroke patients, with more upper outliers with higher BMI values.

4. Diagnostic Analytics tab

1. What is the correlation among features of the datasets?
2. What are relationships among key numerical variables (age, BMI, glucose)?
3. How does the clustering method generate stroke-like and no-stroke-like groups?

Our analysis began with a correlation heatmap, which showed mostly small pairwise correlations (≤ 0.35). Age stood out with notable positive correlations to Hypertension, Heart Disease, and Stroke (≈ 0.25 – 0.30) and a strong negative correlation with Work_Type_children (≈ -0.63), indicating younger participants dominate this category. Other factors like BMI, Glucose, Smoking, Sex, and Residence showed weak associations with Stroke ($\leq \sim 0.13$). For the second question, we used the pairplot showing that age correlates with increases in BMI and glucose, while glucose and BMI have weak relationships, appearing scattered. Some extreme values may indicate outliers or unique metabolic conditions. Overall, the variables are mostly independent, offering distinct insights into stroke risk.

In terms of clustering, K-means ($k = 2$) identified two groups: the "stroke-like" cluster, consisting of 86% of participants with a higher stroke rate of 5.6%, and the "no-stroke-like" group, showing a low rate of 0.3%. Despite some overlapping in PCA projections, the clusters highlight significant differences in health and demographic profiles without using target labels.

5. Predictive Analytics tab

The goal of our machine learning model was to determine whether patients are at risk of a stroke. The relevance of this stroke prediction problem lies in its potential to enhance healthcare outcomes by identifying high-risk individuals early on. The dataset consisted of 10 features, such as Body Mass Index, glucose levels, age, and smoking status. By identifying high-risk patients, clinicians can implement preventive measures, which not only help in resource allocation but also leads to better long-term health outcomes.

6. Prescriptive Analytics tab

We chose to incorporate SHAP (SHapley Additive explanations) to enhance the explainability of our predictive model. To visualize the contribution of each feature to the final prediction, we implemented a waterfall chart, which assigns a quantitative impact value to each feature. This approach allows users to see how individual inputs either increase or decrease the predicted risk. We selected the waterfall chart specifically for its intuitive design, making complex model behavior easier to interpret, especially for users in clinical or decision-making settings.

DESIGN PROCESS

1. Early-stage prototyping

In the early stages of developing the web dashboard, our focus was on creating a clear, intuitive, and user-friendly interface that would allow non-technical users to interact with the stroke prediction results and explore key patterns in the dataset. The following decisions and activities guided our prototyping. The early-stage design prioritized **simplicity, clarity, and usability**. Since the target users include healthcare professionals who may not have technical expertise, the dashboard was designed to communicate results quickly and intuitively, without requiring prior experience in data science.

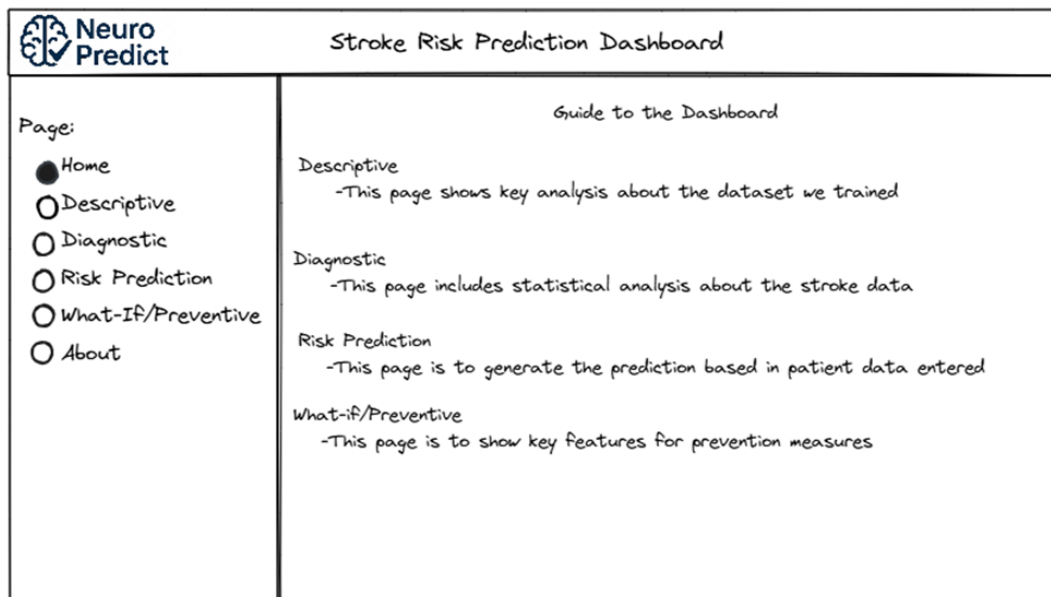


Figure (1): Sample Home page of the dashboard

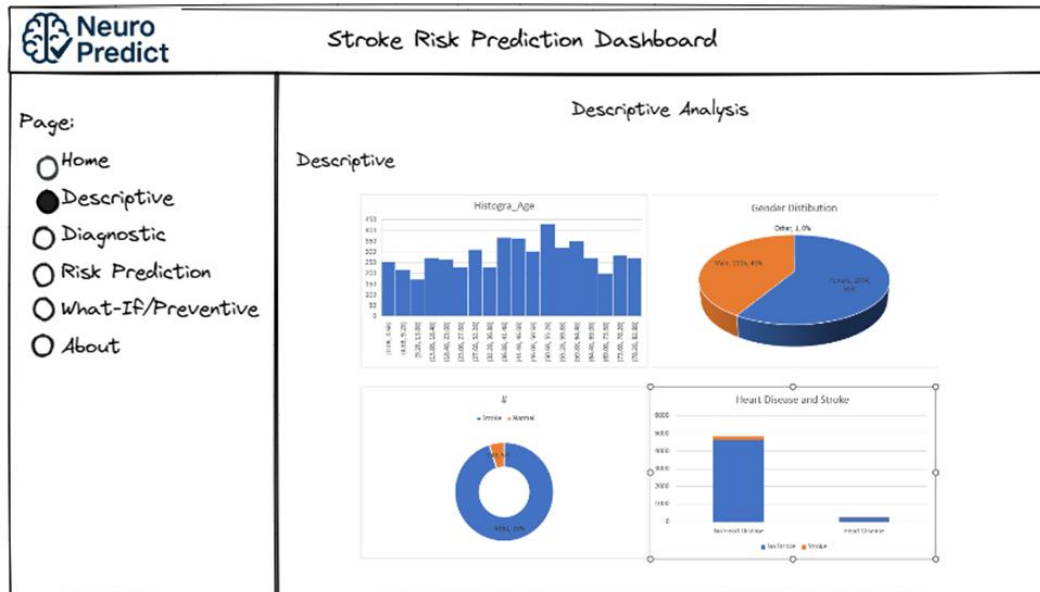


Figure (2): Sample Descriptive page of the dashboard

Figure (3): Sample Risk Prediction page of the dashboard

The goal of our dashboard layout was to make it clear and easy to use, regardless of the individuals' technical skills. One direction we took was to make sure all pages in our dashboard were visible for the target user to be able to switch at will. The pages that are included

- “About”: Is the first page the user will see, and it contains general information about the creators, the dashboard, its purpose, and contact information.

- “Risk Prediction”: The user will go through each query, and the model will calculate the risk and give a preliminary prediction.
- “Preventive Analytics”: Requires the user to go through the risk prediction page, the data is then shown to the user on what influences the risk prediction for that specific user.
- “Descriptive Analytics”: Gives the user the means to inspect our descriptive analytics of the data set used for the project. The user will be able to read the risk factors and demographical findings.
- “Diagnostic Analytics”: In this page, the user will be able to read our diagnostic analysis of our data through different visualization methods, such as heatmaps, plots, and clustering.

The purpose of adding diagnostic and descriptive analytics is to be transparent with our target users. We believe that understanding demographic and diagnostic analysis will help them better understand their own risk prediction.

Low-fidelity prototypes allowed our group to test and visualize our concepts before spending a significant amount of time on them, gave our group a flexible mindset, and made us more open to change and experimentation. Early ideas and prototypes improved our collaboration within our group. We were able to communicate our early ideas and concepts and discussed which methods were preferred and why we wanted the prototypes to be simple and improve on that. Within our group, we kept giving each other feedback on what worked and what did not and documented everything towards the high-fidelity prototype’.

2. First high-fidelity prototype

We chose to conduct user testing after completing the high-fidelity prototype, prioritizing the evaluation of page content and functionality over final design elements such as sidebar colors and information layout, which were still under consideration. Testing was initiated once the machine learning model with optimal performance metrics was integrated and all pages were finalized with the necessary aspects. Due to time constraints, we focused primarily on the risk prediction and preventive tabs, as these were deemed most valuable to end users. Additionally, we ensured that all relevant charts and visualizations, based on the key questions established by the team members from the dataset, were included. After aligning these criteria, the team collectively agreed that the prototype was ready for testing.

Please refer to video demo of the first high fidelity prototype before usability testing

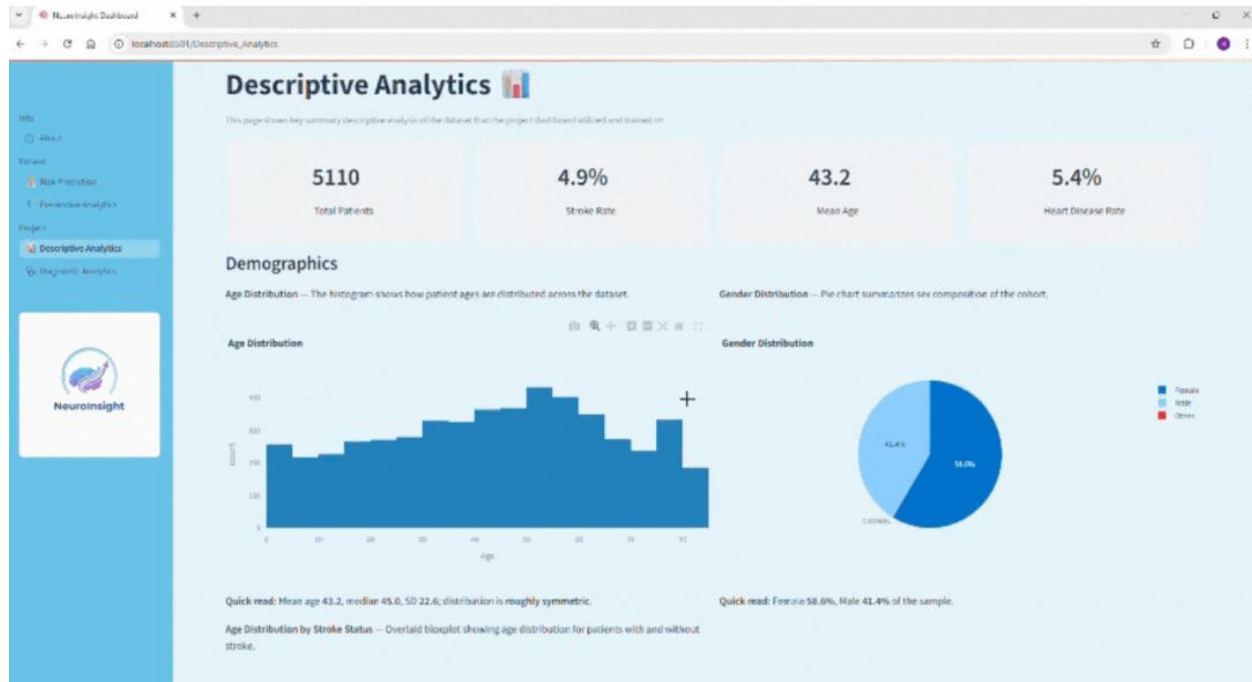


Figure (4) First High Fidelity prototype

USER EVALUATION

One of the primary goals of the Neuroinsight Dashboard is to ensure it is user-friendly for all users, particularly clinicians, who make up our main user group. For user testing, our main objectives are to identify issues that hinder usability, such as navigation errors, and to uncover opportunities for improving the design (4).

We design these questions to answer.

- Is the dashboard intuitive and easy to use for entering data to calculate stroke risk?
- Does the descriptive tab present essential and relevant dataset information in a clear and accessible way?
- Is the Explainable AI component understandable, and do healthcare professionals grasp the key factors influencing the model's decisions?
- Can participants navigate the dashboard smoothly and without confusion?

We also aim to gather overall impressions of the Dashboard and anticipate receiving positive feedback from users. To evaluate usability, we will administer the SUS questionnaire. Additionally, to collect qualitative insights, we will conduct a think-aloud test with three students from the Health Informatics Master's program who have a Healthcare background. The sample size was determined based on time constraints and convenience. Both the questionnaire and the

think-aloud protocol were administered during Sprint 4: Dashboard Interactivity and User Interface Improvement. These sessions were conducted via zoom with all participants to be able to record voice while we observed how the user interacted with the dashboard.

1. System Usability Scale (SUS)

The **System Usability Scale (SUS)** was selected as the quantitative method for evaluating usability. SUS is a validated 10-item questionnaire that uses a five-point Likert scale ranging from *strongly agree* to *strongly disagree*. The questionnaire includes an equal number of positively and negatively worded items (five each), allowing for a balanced assessment of user experience (5). SUS was chosen primarily for its ease of administration and its proven reliability in assessing the usability and user experience of Digital Health applications (6).

Each item is scored on a scale from 0 to 4. For negatively worded items (questions 1, 3, 5, 7, and 9), the score is calculated by subtracting 1 from the scale position. For positively worded items (questions 2, 4, 6, 8, and 10), the score is determined by subtracting the scale position from 5. The total score is then multiplied by 2.5 to yield an overall SUS score ranging from 0 to 100, where higher scores indicate better usability (7).

SUS questionnaire interpretation

There is no established threshold for measuring app acceptability using the SUS. However, some authors suggest that scores above 68 are considered acceptable, while scores below 68 are viewed as below average (6,8), particularly for digital applications.

Participant ID: _____ Site: _____ Date: ____/____/____

System Usability Scale

Instructions: For each of the following statements, mark one box that best describes your reactions to the website *today*.

		Strongly Disagree				Strongly Agree
1.	I think that I would like to use this website frequently.	☐	☐	☐	☐	☐
2.	I found this website unnecessarily complex.	☐	☐	☐	☐	☐
3.	I thought this website was easy to use.	☐	☐	☐	☐	☐
4.	I think that I would need assistance to be able to use this website.	☐	☐	☐	☐	☐
5.	I found the various functions in this website were well integrated.	☐	☐	☐	☐	☐
6.	I thought there was too much inconsistency in this website.	☐	☐	☐	☐	☐
7.	I would imagine that most people would learn to use this website very quickly.	☐	☐	☐	☐	☐
8.	I found this website very cumbersome/awkward to use.	☐	☐	☐	☐	☐
9.	I felt very confident using this website.	☐	☐	☐	☐	☐
10.	I needed to learn a lot of things before I could get going with this website.	☐	☐	☐	☐	☐

Figure (5) System Usability scale sheet example

2. Think-aloud method

For the qualitative insights, we decided to use the Think-aloud technique, where the participants can test the dashboard and verbalize their thoughts while they navigate and use it. Each participant received a task instruction sheet outlining specific actions to perform on the dashboard. Sessions were conducted via zoom in a quiet, controlled environment to minimize distractions and ensure high-quality data collection (9).

Instructions

Welcome and thank you for participating in evaluating the Neuroinsight Dashboard, during this session, we'll ask you to use the dashboard while thinking out loud. This means we'd like you to say everything that goes through your mind as you complete each task, what you're looking at, what you're trying to do, what confuses you, and what feels intuitive. Before we begin, please note that this session will be recorded with your permission. All information you share will be treated as confidential and used solely for the purpose of improving the dashboard's usability. You are free to pause or stop the session at any time. There are no right or wrong answers. We're interested in your thoughts as you interact with the dashboard.

Tasks and Prompts

As you go through each section, please speak aloud your thoughts related to the following:

- 1. Data Entry for Stroke Risk**
 - Is it clear where and how to enter data?
 - Do you feel confident that the inputs are correct?
 - What do you expect to happen after submitting?
- 2. Descriptive Page**
 - What information stands out to you?
 - Is the presentation of the dataset easy to understand?
 - Is anything missing or unclear?
- 3. Explainable AI Section**
 - Do you understand what the model is showing?
 - Can you identify which factors influence the prediction?
 - Would this be useful for a healthcare setting?
- 4. Navigation**
 - Are you able to move between sections easily?
 - Do you ever feel lost or unsure where to click next?
 - What would make navigation smoother?

Lastly, we would like to hear your thoughts on the overall interface or if you have questions, thank you.

3. Results

We administered three questionnaires (see Appendix) and conducted three think-aloud sessions with Health Informatics students from Pharmacy and Medicine backgrounds. To collect responses for the System Usability Scale (SUS) questionnaire, we created a digital form and distributed it via WhatsApp, and we scheduled Zoom meetings based on each participant's availability. All voice recordings were made with prior consent, and participants were clearly informed that their data would be treated confidentially and used exclusively for the purpose of this assignment.

3.1. SUS Results

Table 1. Participants occupation and SUS score

Participant	Occupation	SUS Score
1	Pharmacy	67.5
2	Pharmacy	95
3	General practitioner	75
Average		79

The SUS scores from the three participants varied significantly, resulting in an average score of 79. While this score is considered above average and indicates generally acceptable usability, the variability suggests that user experience is inconsistent. This highlights the need for further improvements, particularly to address the specific needs and expectations of our users.

3.2. Think aloud results:

Task 1: Data Entry for Stroke Risk

Participant One found it clear where data should be entered to generate input, indicating that the overall layout supported task completion. However, it was noted a usability issue with the button colors for selecting options such as Hypertension, Heart Disease, and Married, the use of white made it difficult to distinguish which option was currently selected. Additionally, the work type feature caused confusion due to unclear labeling, leaving the participant uncertain about what each category referred to.

Participant Two noted that the input fields were clearly labeled, which made it easy to understand where and how to enter data. However, he pointed out a usability issue with the button colors,

stating that they were not sufficiently visible. This made it difficult to identify which option was currently selected, potentially affecting the clarity and confidence of user interactions.

Participant Three expressed that manually calculating BMI could be challenging for users and recommended adding input fields for weight and height, allowing the system to automatically compute BMI. While the result was easy to understand, the participant noted that providing a brief explanation of the result would offer more meaningful insights, especially from the perspective of a general practitioner seeking clinical relevance.

Task 2: Descriptive Page

Participant One found this page particularly helpful for understanding the overall characteristics of the dataset, including insights into age distribution and key factors related to stroke. They appreciated the clarity of the visualizations and the explanations provided for each chart, which supported their interpretation of the data. Additionally, they recommended maintaining a consistent color palette, suggesting that if blue is used, it should be applied uniformly across the dashboard to enhance visual coherence and aesthetic appeal.

Participant Two highlighted that key risk factors, particularly age and blood pressure, were clearly emphasized in the visualizations. They found the charts informative and easy to interpret, which supported their understanding of the dataset. However, they suggested including a glossary of terms to enhance clarity, especially for users who may be unfamiliar with specific medical or technical terminology.

Participant Three appreciated the clarity and simplicity of the charts, noting that they were easy to interpret and not overly complex. This design enabled rapid assessment, which is particularly valuable in a clinical setting where time is often limited.

Task 3: Explainable AI Section

Participant One initially expressed confusion about the data presented on the page but later recognized that it represented data from the patient. It was suggested that a brief explanation of the data's purpose and origin would improve clarity for users. Additionally, the participant found the SHAP explanation chart difficult to interpret at first and recommended including supporting text or guidance to clarify its role and relevance in the dashboard.

Participant Two explained that, due to their professional background, it was easy to understand the influential factors in the model. It was added that 'this chart will be helpful for explaining risk to patients in clinical decision-making settings.'

Participant Three also found the information difficult to understand and felt that at least a basic explanation of SHAP was necessary. However, they noted that the chart was helpful in gaining a better understanding of the model.

Task 4: Navigation

Participant One mentioned that the content across all pages was easy to understand. However, they pointed out that labeling pages as 'descriptive' or 'prescriptive' might be confusing for users. Additionally, it was recommended maintaining a consistent color palette throughout the design to support visual coherence.

Participant Two notes that moving between sections is simple and intuitive, and mentioned not feeling lost or unsure on where to click.

Participant Three also found the dashboard easy to navigate. They suggested adding brief explanations or tooltips that appear when hovering over the page tabs, so users can quickly understand the content of each section without needing to click through everything. This would help users locate specific information more efficiently.

Table 2. Tasks and the recommendations for improvements provided by Participants

Task	Recommendation for Improvement
Task 1: Data Entry for Stroke Risk	Improve button color contrast to clearly indicate selected options (Hypertension, Heart Disease, Married). Clarify labeling of work type categories to reduce confusion. Add input fields for weight and height to automatically calculate BMI. Provide brief explanations of the result to enhance clinical relevance.
Task 2: Descriptive Page	Maintain a consistent color palette across the dashboard (uniform use of blue). Include a glossary of terms to clarify medical and technical terminology.
Task 3: Explainable AI Section	Add a brief explanation of the data's origin and purpose to improve clarity. Include supporting text or guidance to help interpret the SHAP chart. Provide a basic explanation of SHAP to improve user understanding.
Task 4: Navigation	Avoid using labels like 'descriptive' or 'prescriptive' for page titles to prevent confusion. Maintain a consistent color palette throughout the design. Add tooltips or hover explanations on page tabs to help users locate information more efficiently.

4. Conclusion and improvements to the final product

Since this usability test could not be conducted with a broader group of healthcare professionals, particularly those without clinical expertise, the results may be subject to bias. However, we found that the usability evaluation, especially the Think-Aloud test, provided valuable insights that significantly improved our dashboard. These findings helped us refine the tool to ensure it is more user-friendly, understandable, and practical. The changes implemented underscore the importance of continuous user feedback, especially in healthcare environments where clarity and usability are essential.

As shown in Table 2, we updated the button colors for Hypertension, Heart Disease, and Married to enhance visual clarity. According to Figure 6, these changes have made it easier for users to identify their selections. Additionally, we replaced the BMI input with separate fields for height and weight, allowing users to enter data directly without needing to calculate BMI themselves.

Risk Prediction

This page predicts stroke risk using the trained model based on the input features below.

Enter Patient Data

Age	45	-	+
Work Type	Private		
Hypertension	☒ No ☐ Yes		
Heart Disease	☒ No ☐ Yes		
Gender	Female		
Ever Married?	☐ No ☒ Yes		
Residence Type	Urban		
Glucose (mg/dl)	91.90	-	+
Height (cm)	150	-	+
Weight (kg)	80.00	-	+
Smoking Status	smokes		

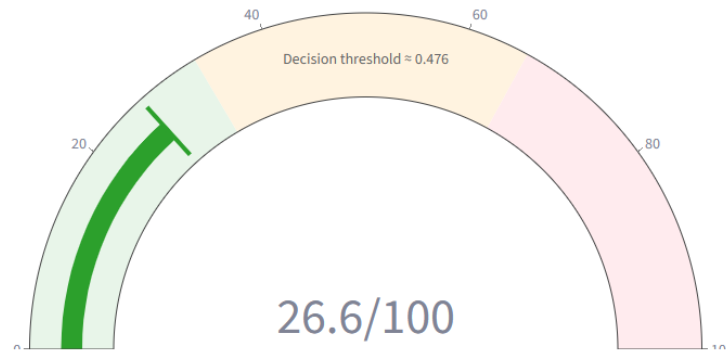
Predict

Figure (6) Risk prediction tab after usability testing

After the dashboard calculates the prediction, we added a risk classification label, such as Low Risk and High risk to help users quickly understand where the patient falls within the risk spectrum. This addition enhances clarity by providing immediate context for the result, making it easier for users to interpret the outcome and take appropriate action.

Model-Estimated Stroke Risk

Low Risk • 27/100



Predicted probability: 0.266

Decision (threshold = 0.476): Low stroke risk (0)

Figure (7) Decision threshold

Preventive Insights

Impact on predicted stroke probability based on patient data using SHAP.

Summary of Patient Input Data

Age 65 Sex Female Married No Hypertension Yes Heart disease Yes BMI 25.5 Glucose 200.0 Residence Urban Work Private Smoking smokes

Top contributors (patient-specific)



How to read this

This chart shows which patient factors most strongly influenced the stroke-risk prediction for this specific person. Each bar represents how much a factor increased or decreased the predicted probability of stroke.

- Red bars → Factors that increase the predicted risk of stroke for this patient.
- Green bars → Factors that reduce the predicted risk of stroke for this patient.
- The longer the bar, the stronger the effect on the prediction.

Example - A patient with Glucose level as 190 mg/dl and BMI as 28

Glucose (+0.070): Having high glucose than normal makes the model to predict this patient with increase probability of stroke.

BMI (-0.047): Not having too high BMI values of the patient reduces the predicted stroke risk slightly for him or her.

Figure (8) Explainability of the stroke prediction model, preventive tab

For the explainability component of the dashboard, we incorporated a clear explanation and guidance to help users interpret the waterfall chart, which visualizes how each feature contributes to the model's prediction. This ensures that users, especially those without technical backgrounds.

can understand the rationale behind the risk assessment. Additionally, we chose to retain the input data from the risk prediction tab alongside the chart, allowing users to easily reference the original patient information while reviewing the explanation. This integration supports transparency and reinforces trust in the system's output.

Regarding color palette, we selected a white and blue color palette for the dashboard to ensure a clean, uniform visual experience. This choice helps avoid overwhelming users with excessive color while maintaining a professional aesthetic. By keeping the design focused and consistent, we aim to enhance usability and ensure that attention remains on the core functionalities and data visualizations.

Declaration of AI tools

We strictly adhered to DSV's AI policy by employing AI tools to assist us in our work. These tools helped us gain a better understanding of Streamlit and supported us in developing our dashboard, particularly with the coding aspects. For debugging, we utilized AI tools to identify errors and implement solutions. Additionally, we used Grammarly in specific sections of this final report to proofread and improve the language flow. We also employed ChatGPT to design our team logo.

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Appendix

Anx (1): SUS Survey form

SUS

B *I* U [↗](#) ~~X~~

Instructions: For each of the following statements, mark one box that best describes your reactions to the website.

I think I would like to use this website

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

I found this website unnecessarily complex

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

I thought this website was easy to use

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

I think that I would need assistance to be able to use this website

	1	2	3	4	5	
Strongly disagree	☐	☐	☐	☐	☐	Strongly agree

3 respuestas

 Vincular con Hojas de cálculo 

Resumen

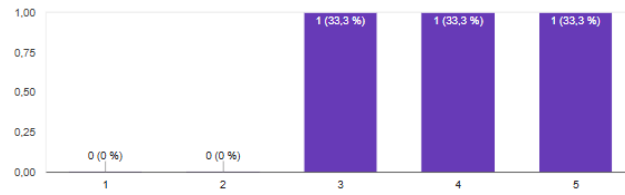
Pregunta

Individual

I think I would like to use this website

 Copiar gráfico

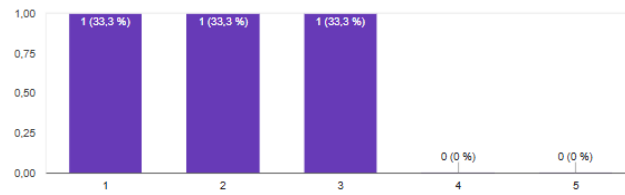
3 respuestas



I found this website unnecessarily complex

 Copiar gráfico

3 respuestas



I thought this website was easy to use

 Copiar gráfico

3 respuestas

1 00

3 respuestas

[Ver en Hojas de cálculo](#)[Resumen](#)[Pregunta](#)[Individual](#)

I think I would like to use this website



1

de 10



I think I would like to use this website

Strongly disagree 1 2 3 4 5 Strongly agree

☐ ☐ ☐ ☐ ☒

1 respuesta

Strongly disagree 1 2 3 4 5 Strongly agree

☐ ☐ ☐ ☒ ☐

1 respuesta

Strongly disagree 1 2 3 4 5 Strongly agree

☐ ☐ ☒ ☐ ☐

1 respuesta

Anx (2) - Evidences of WhatsApp and GitHub Project Groups

